

Spatial Determinants of Copper Infrastructure Theft: An Agent-Based Model with Census-Level Empirical Validation in Southern California

Jiaman Li¹[0009–0009–9322–0400] and Zining Yang²[0009–0007–2491–1625]

¹ George Mason University, Fairfax VA 22030, USA
jli30@gmu.edu

² Southern California Edison, Rosemead CA 91770, USA
zining.yang@sce.com

Abstract. Electric asset vandalism poses persistent public safety and operational risks in Southern California, contributing to multiple electrocution fatalities and service disruptions in recent years. Existing mitigation is largely reactive or is based on costly infrastructure hardening that scales poorly. We present an empirically grounded agent-based model (ABM) for spatially targeted vandalism prevention within Southern California Edison’s (SCE) LA Metro West service area, spanning 842 census tracts. The model binds eight tract-level socioeconomic and environmental variables to space and grounds offender decision-making in four criminological theories: Rational Choice, Routine Activity, Social Learning, and the Resetting (deterrence) effect. Calibrated against California Public Utilities Commission (CPUC) incident data, the model reproduces the observed annual rate of vandalism (190 vs. a 100–400 target) and electrocutions (9.5 vs. a 4–15 target). A Monte Carlo sensitivity analysis shows that annual theft volume rises monotonically with both copper price (a 128% increase across a realistic price range) and the radius of social contagion (a 55% increase), confirming the economic-incentive and social-learning mechanisms respectively. The model’s result concentrates observed vandalism: the top 20% of predicted-risk tracts capture 51.5% of reported incidents (Predictive Accuracy Index = 2.58; AUC = 0.70), with idle commercial accounts the strongest spatial predictor. We make explicit that achievable spatial accuracy is bounded by the sparsity of incident records, and argue the ABM’s distinctive value is its generative, mechanism-transparent dynamics rather than out-predicting regression.

Keywords: Agent-based modeling · Copper theft · Critical infrastructure · Spatial criminology · Social contagion · Empirical calibration.

1 Introduction

Vandalism of electric distribution assets imposes three interacting risks on utilities and the public: direct equipment loss, service interruption, and severe injury or death from contact with energized conductors. In Southern California Edison’s (SCE) territory these incidents occur year over year, and the public-safety

dimension, namely the electrocution of offenders and bystanders, raises the issue above ordinary property crime. Mitigation today is dominated by reactive repair and capital-intensive situational hardening including enclosures, anti-theft coatings, and surveillance [6], both of which scale poorly across a service territory of roughly 50,000 square miles containing about 1.3 million distribution structures.

A more economical posture is spatial targeting: concentrating limited prevention resources where vandalism is most likely to emerge. This requires a model that (i) is grounded in observable neighborhood conditions, (ii) reproduces the real rate and dynamics of incidents, and (iii) yields an actionable ranking of high-risk locations. Prior simulation work on infrastructure vandalism has often relied on aggregated, macro-level parameters (a single region-wide “deprivation” or “price” lever), which falls short of capturing the neighborhood heterogeneity that governs where crime concentrates.

We address this gap with an agent-based model in which each spatial unit carries real world data at the census-tract level, and offender agents make theory-driven decisions conditioned on those local attributes. The model is instantiated for SCE’s LA Metro West area (842 census tracts across Los Angeles and Orange Counties; about 1,057 distribution-circuit segments). Our contributions are:

1. A census-based ABM of electric-asset vandalism that binds eight tract-level variables and incorporates four criminological theories in a single agent decision rule.
2. Calibration to CPUC incident data: the model reproduces observed annual theft and electrocution rates at a realistic scale.
3. A Monte Carlo sensitivity analysis isolating the mechanisms that drive vandalism volume, with copper price as the dominant economic driver and social contagion a strong second.
4. Spatial validation against SCE’s internal incident database using capture-based predictive-policing metrics, with the data-imposed accuracy ceiling made explicit.

2 Related Work

Our work builds directly on a tradition of agent-based, GIS-grounded modeling of techno-social infrastructure. Abdollahian, Yang, and Nelson [1] introduced SEMPro, an agent-based predictive-analytics model of energy infrastructure siting that couples engineering and institutional constraints with GIS land-use layers and U.S. Census residential demographics, and that represents the behavioral diffusion of public support and opposition through interacting agents. Nelson, Cain, and Yang [12] formalized the principle that all politics are spatial, showing how integrating spatially explicit landscape, demographic, and economic data changes the conclusions an agent-based model reaches about local conflict. Wikstrom, Nelson, and Yang [18] then addressed a problem that is often neglected in this literature, namely how to validate an ABM-GIS model against historical spatial data rather than against stylized facts alone. The present study inherits three commitments from this line of work: agent decisions bound to real

census geography, explicit representation of behavioral diffusion, and validation against observed spatial outcomes. It extends them to a domain they have not previously addressed, the prediction of infrastructure crime.

A second and rapidly expanding body of work applies agent-based simulation to crime, much of it organized around environmental criminology and, in particular, Routine Activity Theory, with offenders, targets, and guardians moving through an urban environment. Recent theory-guided agent-based models couple criminological theory with real crime, environmental, and census data to reproduce the spatiotemporal distribution of offenses and to support policing decisions [11]; comparative studies relate agent-based and space-syntax simulations of pedestrian movement and visibility to forecast crime patterns [10]; and generative-agent frameworks drive offender routines with empirically calibrated mobility models [21]. These models demonstrate the value of mechanism-based simulation for general urban crime, but they are seldom applied to the theft of physical infrastructure, where the targets are fixed utility assets rather than people or premises and where the economic driver of resale value is central.

In parallel, statistical and deep-learning methods have become the default tools for crime-hotspot prediction. Recent architectures combine residual and recurrent networks to forecast hotspots from historical incident sequences [15], and multi-scale neural networks map hotspots at fine spatial resolution [9]. Such methods can achieve strong predictive fit where incident data are dense, but sparse, zero-inflated spatial counts remain challenging even for purpose-built architectures [9], precisely the regime of rare infrastructure-crime records that we encounter and quantify. They are also correlational: trained on past outcomes, they cannot answer counterfactual questions about conditions that have not yet occurred, such as a sustained copper-price shock or a change in the offender population. Our agent-based approach is deliberately complementary, trading a measure of raw predictive fit for transparency of mechanism and the ability to run policy-relevant scenarios.

Finally, a more applied literature documents the infrastructure-crime problem itself. Sidebottom et al. [16] empirically showed that copper theft tracks the price of copper in volatile metal markets, a relationship our model encodes and reproduces. Situational crime-prevention research [6] frames the hardening measures that utilities deploy, while recent industry assessments report a sharp rise in theft and vandalism incidents across communications and transportation infrastructure [14]. Predictive-policing evaluation has, in turn, standardized capture-based metrics such as the Predictive Accuracy Index for sparse incident data [5], which we adopt. What is missing is a generative, mechanism-transparent model that links these socioeconomic drivers to observed spatial outcomes, which is the gap this paper addresses.

3 Model

The model is implemented in NetLogo [19,20] with its GIS extension. The source and data are available from the authors.

Table 1. Tract-level variables (U.S. Census ACS 2020 and utility records).

Field	Meaning	Theoretical role
MHI20	Median household income	Relative deprivation / poverty
IDLE_CUST	Idle commercial accounts (≥ 1 yr)	Suitable target / weak guardianship
TRANSIT	Transit stops	Accessibility / escape routes
PCT_COMM	Percent commercial zoning	Target richness
RISERS	Riser/transformer asset count	Asset exposure
CRV	Recycling centers (CalRecycle)	Fence availability
TOTAL_POP	Total population	Exposure scaling
Internal	Utility internal vandalism reports	Validation target (not in decision rule)

3.1 Environment and Data

The spatial component is a matrix of patches onto which tract polygons are projected via areal coverage; each patch inherits the eight attributes of its underlying tract(s). Distribution-circuit geometry is loaded from SCE’s public shapefiles, and electric asset agents (transformers) are placed along these circuits. Table 1 lists the eight tract variables and their theoretical roles. Data integrity was verified end-to-end: all eight fields match between the source records and the model shapefile across 842 tracts (99.8% areal coverage; two offshore islands excluded).

Although the data provide real circuit geometry, point-level transformer coordinates are not available from public data source. We therefore place asset agents uniformly at random along the real SCE circuit network; when the asset count exceeds the number of circuit cells, multiple assets share a cell (sampling with replacement), allowing the asset base to be scaled toward the territory’s true magnitude without being capped by raster resolution. The tract **RISERS** count is an aggregate that is biased toward commercially classified tracts and is therefore not used as a placement prior. This conservative choice isolates socioeconomic rather than asset-density effects on where vandalism concentrates.

3.2 Agents and the Decision Rule

Offender agents carry a personal income (drawn from a realistic distribution), a crime-history counter, a decaying social-contagion term, and a stable skill trait $\sim U(0.60, 0.80)$ representing baseline competence. One step represents one calendar day. On each step an offender near an active asset forms an incentive:

$$\begin{aligned}
incentive = & (copper - 3.0)/8 + (inflation/100) \cdot 0.3 \\
& + \max(0, (MHI - income)/10^5) \cdot 0.2 + history \cdot 0.03 \\
& + (80000 - MHI)/(2 \times 10^5) + transit/500 + idle/200 \\
& + (transit \cdot idle)/5000 + comm/100 + CRV/100 + contagion, \quad (1)
\end{aligned}$$

where the market, inflation, and relative-deprivation terms are doubled when a step occurs at night.

Table 2. Calibrated annual outputs (mean over replicate runs).

Output	Model CPUC target range		Status
Thefts / year	190	100–400	within range
Electrocutions / year	9.5	4–15	within range

The vandalism decision is probabilistic: an agent attempts an offense with probability $P = \text{attempt_rate} \cdot \text{incentive}$. A probabilistic rule is essential: under a hard threshold the composite incentive saturates above the cut-off, rendering the economic and social mechanisms inert (we observed flat sensitivity sweeps under a threshold rule). The single scalar *attempt_rate* is the calibration knob for baseline vandalism frequency.

On an attempt, the theft succeeds with a heterogeneous probability equal to the agent’s skill plus an experience bonus (+0.01 per prior offense, capped at +0.15; total capped at 0.95), replacing a fixed success rate and letting individual heterogeneity express itself. A successful theft inactivates the asset, raises crime history, and propagates a contagion boost to offenders within the contagion radius; with probability 0.05 it triggers an electrocution. This 0.05 is the empirical ratio of electrocutions to thefts in CPUC/SCE records (about 14 electrocutions per 330 annual thefts ≈ 0.043 , rounded). On failure, most agents are deterred (incentive reduced, long cooldown) and a minority reset (short cooldown), per the Resetting effect. Inactivated assets recover stochastically.

3.3 Calibration

The simulation runs 365 steps, capturing a full year’s dynamics. The two unanchored parameters, namely the offender population and the attempt rate, are calibrated so that annual theft and electrocution counts fall within CPUC-derived target ranges [4]. We scale to 2,500 asset agents and 1,000 offender agents to approximate the territory’s operational magnitude, and calibrate *attempt_rate* = 0.0015. All reported results use Monte Carlo replication via NetLogo BehaviorSpace.

4 Experiments and Results

4.1 Rate Calibration against CPUC

At the calibrated configuration the model reproduces observed incident rates (Table 2). Both outputs fall comfortably within the CPUC target ranges. Because electrocutions equal the fixed empirical ratio times thefts, sensitivity to that ratio is exactly linear and is fully captured by the theft-rate analysis; we therefore do not sweep it separately.

Table 3. Theft volume versus copper price (calibrated baseline).

Copper price (\$/lb)	Thefts / year	Electrocutions / year
3.5	119	8.0
4.5	191	12.3
5.5	272	14.0

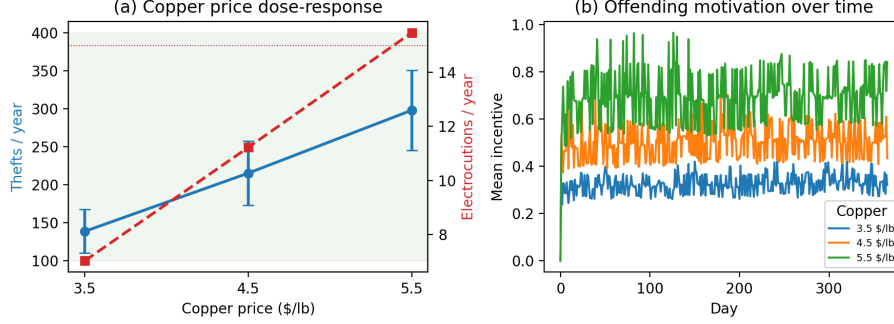


Fig. 1. Economic mechanism. (a) Copper price drives annual theft volume (blue, left axis) and electrocutions (red, right axis); the green band marks the CPUC theft range. (b) Population-mean offender motivation over the 365-day horizon, stratified by copper price, showing higher and escalating motivation at high price.

4.2 Mechanism Sensitivity

Copper price (economic mechanism) Sweeping the copper price across a realistic range creates a strong, monotonic dose-response in theft volume (Table 3): a price increase from \$3.5 to \$5.5/lb raises annual thefts by 128%, well above replicate noise, while remaining inside the CPUC range throughout. The population mean incentive rises with price and, at the high-price end, escalates over the simulated year (from ≈ 0.47 to ≈ 0.84) as crime history and contagion reinforce one another. This emergent temporal dynamic is one that a static specification cannot produce (Fig. 1). This is the model’s clearest mechanism result and direct evidence for the Rational-Choice channel, consistent with empirical price–theft elasticity in metal markets [16].

Full-factorial sensitivity A full-factorial design over copper price, inflation, and contagion radius (60 parameter combinations, Monte Carlo replicated, 300 runs in total) ranks the drivers of theft volume by marginal effect (Fig. 2). Copper price is the most significant variable, spanning a range of 159 thefts per year across the tested prices. The contagion radius is a strong second driver, spanning a range of 99 thefts per year, while inflation contributes a smaller effect of 24 thefts per year. The substantial contagion effect is noteworthy: it confirms that the Social Learning mechanism materially shapes outcomes once offender

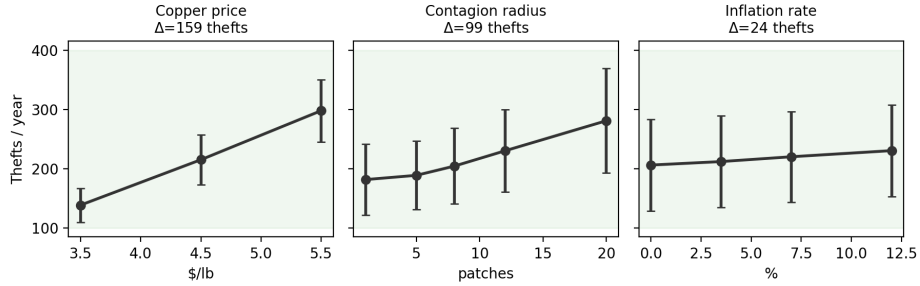


Fig. 2. Full-factorial sensitivity of annual theft volume to copper price, contagion radius, and inflation (marginal effects, with standard deviation bars over replicates; green band marks the CPUC range). Copper price is the dominant driver and social contagion a strong second.

Table 4. Spatial risk ranking versus observed vandalism.

Predicted top-risk tracts	Captured share of vandalism	PAI
Top 5%	20.3%	4.06
Top 10%	28.9%	2.89
Top 20%	51.5%	2.58
Top 30%	62.4%	2.08

density is realistic, which distinguishes the ABM from a purely economic account and motivates its place at a social-modeling venue. All marginal responses are monotonic, and the baseline configuration remains within the CPUC ranges across the interior of the grid.

4.3 Spatial Validation against internal database

We rank tracts by model-derived risk and compare against the SCE internal vandalism database. Because incident records are sparse, with only 144 of 842 tracts (17%) reporting any vandalism, we adopt predictive-policing metrics (capture rate and the Predictive Accuracy Index, PAI [5]) rather than raw correlation. The top 20% of predicted-risk tracts capture 51.5% of all reported incidents (Table 4; PAI = 2.58; discrimination AUC = 0.70), squarely within the PAI 2–5 range reported in the predictive-policing literature. Idle commercial accounts are the strongest single spatial predictor.

We emphasize an honest bound: since the internal database is a sparse count, any model using these eight covariates, whether an ABM, a regression, or a machine-learning method, attains at most a tract-level rank correlation of $\rho \approx 0.29$ with observed vandalism. The model’s risk surface reaches $\rho \approx 0.26$, essentially at this ceiling. Modest absolute spatial correlation is therefore a property of the data, not a model deficiency, which is precisely why we report capture/PAI rather than raw correlation.

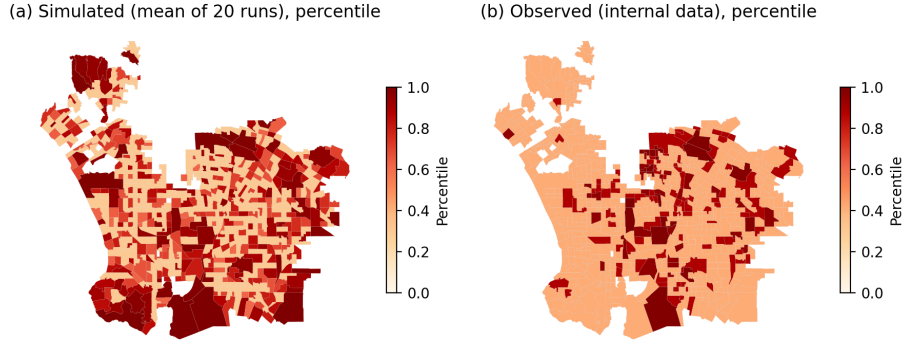


Fig. 3. Spatial comparison of predicted and observed vandalism, both shown as tract percentile ranks. (a) Simulated theft density averaged over 20 baseline replicates; (b) observed incidents (utility internal data). Both concentrate in the central and southern corridor of the service area; the quantitative overlap is reported in Table 4.

5 Discussion

ABM Contributions Since spatial fit is data-capped, the ABM’s value is not out-predicting cross-sectional regression on a single map. Its value is generative: it reproduces the real rate of incidents (Sec. 4.1), responds mechanistically to economic and social drivers (Sec. 4.2) including emergent temporal escalation, and provides a substrate for counterfactual experiments (price shocks, targeted hardening) that a statistical or machine-learning model cannot offer. The model is best read as a calibrated, mechanism-transparent risk generator rather than a black-box classifier.

Limitations (1) Incident sparsity caps achievable spatial accuracy ($\rho \approx 0.29$). (2) Point-level asset locations are unavailable; we place assets along real circuits but uniformly, understating asset heterogeneity. (3) The raster resolution is coarser than the tract layer, introducing aggregation loss in tract-level validation. (4) Internal data captures reported rather than all vandalism and is an imperfect ground truth.

Broad Application The framework, which binds real spatial covariates to agents, grounds decisions in theory, calibrates to incident rates, and validates with capture and PAI metrics, transfers directly to other distributed-infrastructure crimes such as those targeting water, telecommunications, and transportation networks.

6 Conclusion

We presented a census-based agent-based model of electric-asset vandalism for SCE’s 842-tract LA Metro West area. The model reproduces CPUC-observed vandalism and electrocution rates, exhibits strong economic and social-contagion mechanisms with emergent temporal escalation, and ranks spatial risk such that the top 20% of tracts capture over half of observed incidents ($\text{PAI} = 2.58$).

The broader contribution is what a generative, agent-level model offers that a purely statistical or machine-learning predictor does not. First, it stays closer to the real social process: every agent carries individual heterogeneity in skill, criminal history, income, and local exposure, so aggregate outcomes emerge from many distinct decisions rather than from a fitted average. Second, it remains predictive in an operational sense, producing a transparent risk ranking that concentrates real incidents and can guide where to act. Third, and most distinctively, it enables scenario analysis. Since behavior is generated from mechanisms rather than from correlations in historical data, the model can answer counterfactual questions such as the effect of a copper-price shock, an inflation shock, or a wider radius of social contagion, none of which a model trained only on past outcomes can address.

These properties point to a clear next step. Since the model localizes risk and already represents guardianship explicitly, it is a natural test bed for simulating mitigation strategies. Future work will reintroduce mitigation mechanisms including target hardening, enclosures, surveillance, and patrol, and evaluate mitigation strategies in simulation, for example, by hardening the highest-risk tracts and measuring the resulting reduction in vandalism and electrocutions before any field deployment. Coupling validated risk prediction with mechanism-based intervention testing is where a theory-grounded ABM contributes most to infrastructure security planning.

Disclosure of Interests. authors have no competing interests to declare that are relevant to the content of this article.

References

1. Abdollahian, M., Yang, Z., Nelson, H.: Techno-social energy infrastructure siting: Sustainable Energy Modeling Programming (SEMPPro). *J. Artif. Soc. Soc.Simul.* **16**(3), 6 (2013)
2. Akers, R.L.: Rational choice, deterrence, and social learning theory in criminology. *J. Crim. Law Criminol.* **81**(3), 653–676 (1990)
3. Bandura, A.: *Social Learning Theory*. Prentice-Hall, Englewood Cliffs (1977)
4. California Public Utilities Commission: Electric Safety and Reliability Branch incident data, 2023–2024
5. Chainey, S., Tompson, L., Uhlig, S.: The utility of hotspot mapping for predicting spatial patterns of crime. *Secur. J.* **21**, 4–28 (2008)
6. Clarke, R.V.: *Situational Crime Prevention: Successful Case Studies*. Harrow and Heston, Guilderland (1997)

7. Cohen, L.E., Felson, M.: Social change and crime rate trends: a routine activity approach. *Am. Sociol. Rev.* **44**(4), 588–608 (1979)
8. Cornish, D.B., Clarke, R.V.: *The Reasoning Criminal: Rational Choice Perspectives on Offending*. Springer, New York (1986)
9. Jing, C., Lv, X., Wang, Y., Qin, M., Jin, S., Wu, S., Xu, G.: A deep multi-scale neural network for crime hotspot mapping prediction. *Comput. Environ. Urban Syst.* **109**, 102089 (2024)
10. Mara, F., Altafini, D., Cutini, V., Malleson, N.: Simulation to forecast crime patterns: comparing space syntax and agent-based models in exploring pedestrian movement and visibility. *Environ. Plan. B Urban Anal. City Sci.* **52**(9), 2171–2195 (2025)
11. Moradi, S., Zhou, Y., Gullion, C.L., Rosenberger, J.M., Chen, V.C.P., Kan, C.: Theory-guided agent-based modeling for crime prediction and decision support. *Expert Syst. Appl.* **296**, 129181 (2026)
12. Nelson, H.T., Cain, N.L., Yang, Z.: All politics are spatial: integrating an agent-based model with spatially explicit landscape data. In: Campbell, H., Kim, Y., Eckerd, A. (eds.) *Rethinking Environmental Justice in Sustainable Cities: Insights from Agent-Based Modeling*. Routledge, London (2015)
13. Pogarsky, G., Piquero, A.R.: Can punishment encourage offending? Investigating the resetting effect. *J. Res. Crime Delinq.* **40**(1), 95–120 (2003)
14. Protect CCI: Critical infrastructure attacks report: protecting the nation’s communications infrastructure from theft and vandalism. Industry report (2025). <https://protectcci.org>
15. Shahmoradi, N., Alesheikh, A.A., Jafari, A., Lotfata, A.: Hybrid ST-ResNet and LSTM approach for precise crime hotspot prediction. *Sci. Rep.* **15**, 40754 (2025)
16. Sidebottom, A., Belur, J., Bowers, K., Tompson, L., Johnson, S.D.: Theft in price-volatile markets: on the relationship between copper price and copper theft. *J. Res. Crime Delinq.* **48**(3), 396–418 (2011)
17. Sutherland, E.H.: *Principles of Criminology*, 4th edn. Lippincott, Philadelphia (1947)
18. Wikstrom, K., Nelson, H., Yang, Z.: Agents in space: validating ABM-GIS models. In: *AHFE International Conference*. Springer (2019)
19. Wilensky, U.: *NetLogo*. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston (1999). <http://ccl.northwestern.edu/netlogo/>
20. Wilensky, U., Rand, W.: *An Introduction to Agent-Based Modeling*. MIT Press, Cambridge (2015)
21. Zeng, Q., Zhao, R., Mao, J., Li, H., Xu, F., Li, Y.: CrimeMind: simulating urban crime with multi-modal LLM agents. *arXiv:2506.05981* (2025)